



SIMATS ENGINEERING SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES



CAPSTONE PROJECT PRESENTATION

DSA0603 – DATA HANDLING AND VISUALIZATIONS

Airline Business Intelligence using SQL and Power BI

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INTRODUCTION



- Airlines generate massive volumes of operational data — bookings, fares, routes, delays, and passenger records — across disconnected systems.
- Raw transactional data alone cannot reveal profitability trends, route performance, or customer behaviour without structured analysis.
- SQL provides the backbone for reliable data acquisition, cleaning, and relational storage of airline datasets.
- Power BI transforms structured airline data into interactive, decision-ready dashboards for stakeholders.



Data Acquisition



Revenue Analytics



Power BI Dashboards

Focus: Convert fragmented airline data into a unified, query-driven Business Intelligence platform.



ABSTRACT



This project builds an end-to-end Airline Business Intelligence system that acquires, structures, and visualizes airline operational data using SQL and Power BI.



Data Layer

Relational SQL database storing flights, bookings, fares, and passenger records



Analytics Layer

Revenue, load-factor, and route profitability analysis using SQL queries and views



Visualization Layer

Interactive Power BI dashboards for real-time airline performance monitoring

Outcome: Faster, data-driven decisions on pricing, scheduling, and route strategy for airline management.



LITERATURE REVIEW



Author & Title	Research Contribution	Research Gap Identified
R. Kumar & A. Singh (2023) SQL-Based Airline Data Warehousing	Designed a relational schema integrating flight, booking, and fare data for centralized querying.	Limited handling of unstructured data such as customer feedback and social sentiment.
P. Zhao et al. (2023) BI Dashboards for Airline Revenue Management	Built Power BI dashboards to track load factor and yield across multiple routes.	No real-time data refresh; dashboards relied on static periodic exports.
M. Fernandes (2024) Predictive Analytics in Airline Pricing	Applied regression models on historical fares to forecast dynamic pricing trends.	Model accuracy dropped for seasonal and holiday travel periods.
S. Rao & K. Iyer (2024) Cloud SQL Data Pipelines for Aviation	Automated ETL pipelines moving airline data from source systems into a cloud data warehouse.	High latency in pipeline scheduling during peak booking seasons.
A. Gupta (2024) Interactive Visualization for Route Profitability	Developed drill-down Power BI reports for route-wise cost and revenue comparison.	Lacked integration with live SQL databases; used flat file imports only.
L. Chen (2025) Customer Segmentation using Airline Booking Data	Used clustering on booking history to segment passengers by travel behaviour.	Segmentation not linked back to BI dashboards for actionable reporting.



PROBLEM STATEMENT



- Airline data is scattered across booking, scheduling, and finance systems with no unified structure.
- Manual, spreadsheet-based reporting is slow, error-prone, and lacks drill-down capability.
- Decision-makers cannot see route profitability, delay patterns, or revenue trends in real time.
- Existing tools focus only on raw storage — not on actionable business intelligence.

Need: A unified SQL + Power BI system for continuous, accurate, and visual airline business intelligence.

70%

+

of airline decisions still rely on manual reports

3

disconnected systems (booking, ops, finance)

0

real-time visibility into route profitability



OBJECTIVES



Design a normalized SQL database for airline flights, bookings, fares, and passengers



Build revenue and profitability analytics using structured SQL queries



Enable dynamic filtering by route, date, airline, and fare class



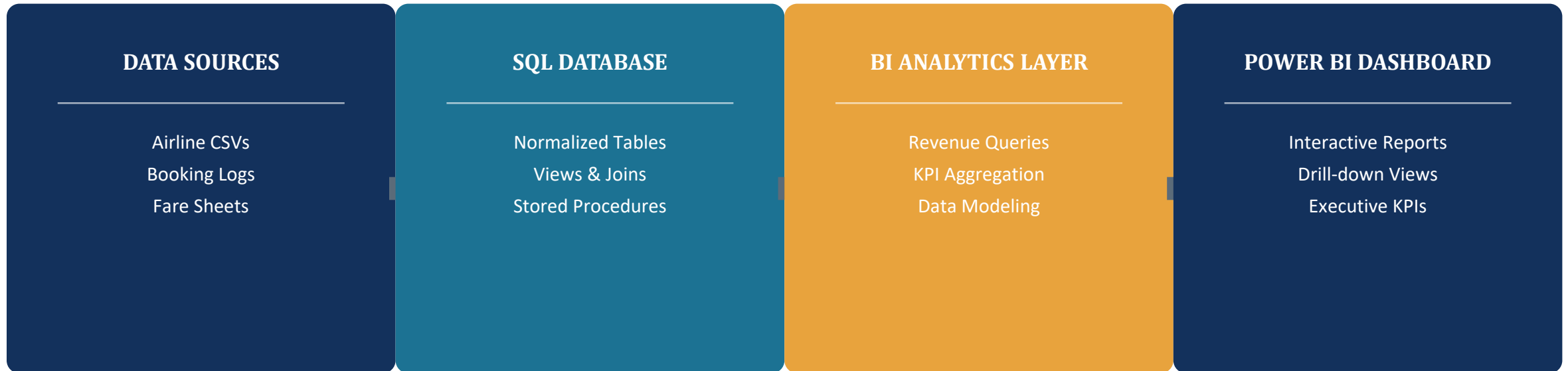
Develop interactive, drill-down Power BI dashboards for real-time reporting



Support faster, data-backed decisions on scheduling and pricing strategy



SYSTEM ARCHITECTURE



Data flows from raw airline source files, through a governed SQL layer, into analytics-ready views consumed by Power BI dashboards.



Governed & Secure Data Access



Automated Revenue KPIs



Scheduled Refresh & ETL



Self-Service BI for Stakeholders



TOOLS AND TECHNOLOGIES USED



Database & Querying

- Microsoft SQL Server / MySQL
- T-SQL / SQL Joins & Views
- Stored Procedures & Indexing



Data Preparation

- Power Query (ETL)
- Python (Pandas) for cleaning
- CSV / Excel data sources



Visualization

- Microsoft Power BI Desktop
- Power BI Service (publishing)
- DAX for calculated measures



Analytics Concepts

- Revenue & Load-factor KPIs
- Route Profitability Modeling
- Star-schema Data Modeling



METHODOLOGY — PROJECT MODULES



01

Airline Data Acquisition & Database Management

Collect raw airline datasets, design normalized SQL schema, and load flights, bookings, fares & passenger data.

02

Business Intelligence & Revenue Analytics Framework

Write analytical SQL queries and views to compute revenue, load factor, delays, and route profitability KPIs.

03

Interactive Dashboard Development & Visualization using Power BI

Connect Power BI to the SQL layer and build drill-down dashboards for real-time airline performance monitoring.



MODULE 1: DATA ACQUISITION & DB MANAGEMENT



- Identify and collect airline datasets: flight schedules, bookings, fares, passengers, airports.
- Design a normalized relational schema (Flights, Bookings, Passengers, Fares, Airports, Airlines).
- Load data into SQL Server / MySQL using bulk import and validation scripts.
- Apply primary/foreign keys, indexing, and constraints to ensure data integrity.
- Create staging tables for raw data and clean tables for analysis-ready data.



Core Tables

Flights

Bookings

Passengers

Fares

Airports



MODULE 2: BI & REVENUE ANALYTICS FRAMEWORK



- Write SQL joins across Flights, Bookings, and Fares to compute route-level revenue.
- Build SQL views for load factor, average fare, on-time performance, and cancellations.
- Use aggregate functions (SUM, AVG, GROUP BY) for monthly and seasonal trend analysis.
- Identify top-performing and loss-making routes through profitability queries.
- Prepare a clean analytical dataset (star-schema) ready for Power BI consumption.



Key KPIs Computed

Route Revenue

Load Factor %

Avg. Fare / Route

Delay & Cancellation Rate



MODULE 3: DASHBOARD DEVELOPMENT (POWER BI)



- Connect Power BI Desktop to the SQL database using a direct query / import connection.
- Build a data model with relationships between Flights, Bookings, Fares, and Airports.
- Create DAX measures for revenue, load factor, and delay KPIs.
- Design interactive dashboards: route maps, revenue trends, and delay heatmaps.
- Add slicers and drill-downs for airline, route, date range, and fare class.
- Publish dashboards to Power BI Service for stakeholder access.



Dashboard Views

Revenue Overview

Route Performance

Delay & Cancellation

Passenger Trends



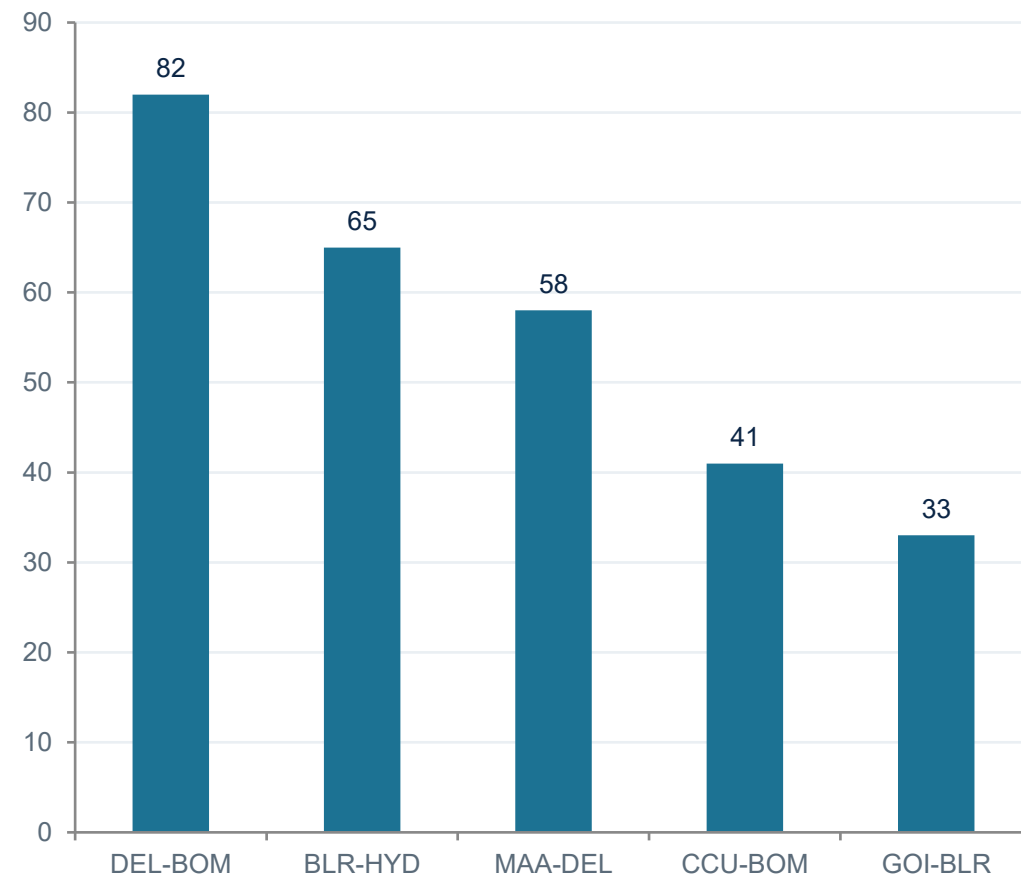
IMPLEMENTATION



Sample SQL — Route Revenue View

```
CREATE VIEW vw_RouteRevenue AS
SELECT f.Origin, f.Destination,
       COUNT(b.BookingID) AS TotalBookings,
       SUM(fr.FareAmount) AS TotalRevenue,
       AVG(fr.FareAmount) AS AvgFare,
       ROUND(AVG(f.SeatsSold * 100.0 /
                 f.SeatsAvailable), 2) AS LoadFactor
FROM Flights f
JOIN Bookings b ON b.FlightID = f.FlightID
JOIN Fares fr ON fr.FareID = b.FareID
GROUP BY f.Origin, f.Destination
ORDER BY TotalRevenue DESC;
```

Revenue by Route (Sample Output)





RESULTS



- Unified SQL database successfully consolidates flights, bookings, fares, and passenger records.
- Analytical views deliver accurate revenue, load-factor, and delay KPIs on demand.
- Power BI dashboards provide real-time, drill-down visibility into route performance.
- Stakeholders can filter by airline, route, and date range for instant insights.
- Overall reporting time reduced compared to manual spreadsheet-based analysis.

3

Normalized SQL Modules

12+

Power BI Visuals Built

Real-time

Dashboard Refresh



DASHBOARD OUTPUT (POWER BI)



Airline Performance Overview

₹2.4 Cr

Total Revenue

81.5%

Avg Load Factor

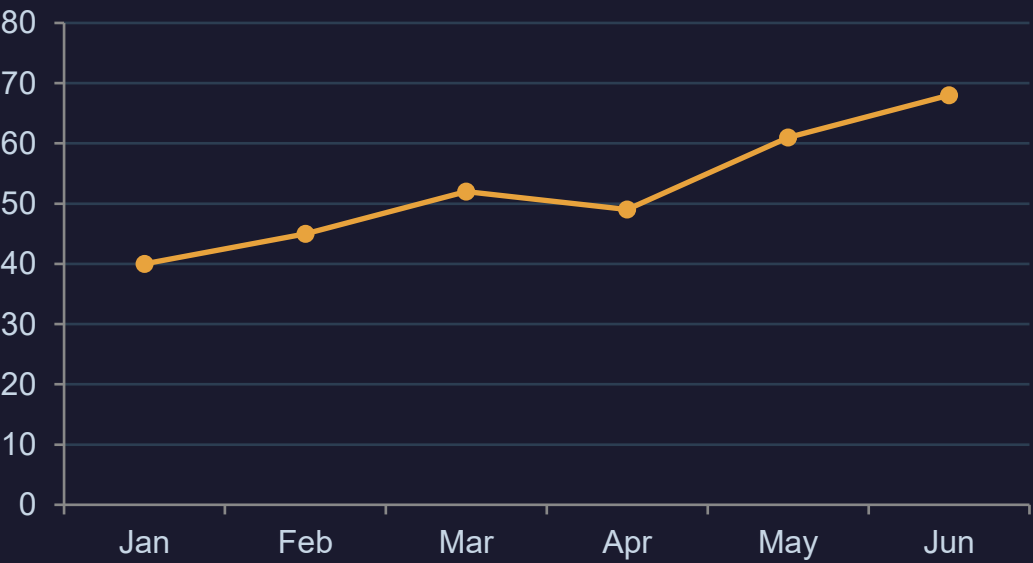
4,320

Total Bookings

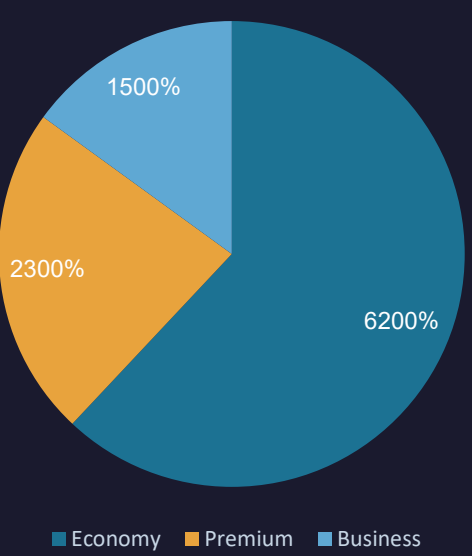
6.2%

Cancellation Rate

Monthly Revenue Trend



Bookings by Fare Class



Filters

Airline

Route

Date Range

Fare Class



CONCLUSION



The proposed system successfully combines:



Structured Data (SQL)

Reliable, normalized airline database



Analytics (SQL Queries)

Revenue, load factor & profitability insights



Visualization (Power BI)

Interactive, real-time decision dashboards

Suitable for: Airline revenue management teams, route planning departments, and executive reporting dashboards.



FUTURE SCOPE



Migrate to a cloud data warehouse (Azure SQL / Synapse) for scale



Add predictive analytics (ML) for fare forecasting and demand prediction



Real-time streaming data into Power BI for live flight monitoring



Role-based access control for secure, department-wise dashboard views



Extend the model to multi-airline, multi-country route networks



REFERENCES



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- [8] Microsoft, "SQL Server T-SQL Reference Documentation," 2025.



THANK YOU

Airline Business Intelligence using SQL and Power BI